

1. Use Gaussian elimination and pivoting technique to solve

$$1.19x_1 + 2.11x_2 - 100x_3 + x_4 = 1.12$$

$$14.2x_1 - 0.112x_2 + 12.2x_3 - x_4 = 3.44$$

$$100x_2 - 99.9x_3 + x_4 = 2.15$$

$$15.3x_1 + 0.110x_2 - 13.1x_3 - x_4 = 4.16$$

1. Solution using Gaussian elimination with pivoting:
 $x_1 = 0.1768, x_2 = 0.0127, x_3 = -0.0207, x_4 = -1.1833$

2. Find the inverse of the matrix A where

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$$A = \begin{bmatrix} 4 & 1 & -1 & 0 \\ 1 & 3 & -1 & 0 \\ -1 & -1 & 6 & 2 \\ 0 & 0 & 2 & 5 \end{bmatrix}$$

2. Inverse of matrix A:
 $\begin{bmatrix} 0.27969349 & -0.08045977 & 0.03831418 & -0.01532567 \\ -0.08045977 & 0.37931034 & 0.05747126 & -0.02298851 \\ 0.03831418 & 0.05747126 & 0.21072797 & -0.08429119 \\ -0.01532567 & -0.02298851 & -0.08429119 & 0.23371648 \end{bmatrix}$

3. Use Crout factorization for a tri-diagonal system to solve the problem

$$\begin{bmatrix} 3 & -1 & 0 & 0 \\ -1 & 3 & -1 & 0 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -1 & 3 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{Bmatrix} = \begin{Bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{Bmatrix}.$$

3. Solution of the tridiagonal system using Crout factorization:
 $x_1 = 1.4364, x_2 = 2.3091, x_3 = 2.4909, x_4 = 1.1636$